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Robust Co-planning of distributed photovoltaics and energy storage for enhancing the hosting capacity of active distribution networks

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ABSTRACT

The inherent uncertainty of photovoltaic systems (PVs) combined with the limited hosting capacity of conventional distribution networks constrains accessible PV capacity, consequently reducing both economic efficiency and voltage stability. To address these challenges, this study proposes an integrated co-planning framework that explicitly incorporates PV uncertainty via a distributionally-robust optimization model designed to enhance grid hosting capacity. The proposed framework systematically embeds short-circuit current constraints to co-optimize economic benefits and operational security. The methodology employs a two-stage optimization approach. First, a distributionally-robust optimization algorithm constructs a probability distribution set for PV uncertainty to improve scenario representation. Subsequently, a second-order cone programming (SOCP) model is formulated for coordinated PV and the energy storage system (ESS) deployment under multiple scenarios, aiming to minimize total economic costs and voltage deviations. Notably, the model incorporates dynamic short-circuit current constraints at grid connection points, in which PV uncertainty is accounted for through iterative calculations using an equivalent voltage source method. To enhance computational efficiency, an improved column-and-constraint generation (C&CG) algorithm is developed by integrating cutting planes derived from short-circuit current validation results. Case studies conducted on a 43-bus distribution network in Zhejiang Province demonstrate that the proposed method achieves a 60% increase in permissible PV capacity and a 16.7% reduction in voltage deviation compared to conventional approaches without ESS coordination or those using decoupled planning strategies. These results validate the effectiveness of integrated short-circuit current constraints in facilitating high renewable penetration while maintaining grid operational security.

1. Introduction

The large-scale integration of distributed photovoltaic (PV) systems with high uncertainty, has increasingly strained the hosting capacity of existing distribution infrastructure. This constraint not only limits PV penetration levels but also degrades economic efficiency, thus creating bottlenecks for renewable energy integration. In active distribution networks, hosting capacity is quantified as the maximum permissible renewable generation capacity while maintaining operational security metrics (e.g., short-circuit current, voltage deviation), within allowable thresholds [1]. Structural weaknesses in existing networks, compounded by uneven load distribution, frequently lead to line congestion and system imbalance, further restricting renewable integration. Consequently, developing systematic methodologies for hosting capacity enhancement emerges as a critical research priority for achieving high renewable penetration.

Energy storage systems (ESSs), as flexible resources, play a pivotal role in high-renewable-penetration networks. ESSs mitigate PV output volatility through dual-mode regulation (power smoothing and temporal energy shifting), enabling economically viable planning solutions [2].Existing research primarily focuses on voltage regulation optimization [3,4] andcost minimization [5,6]. For example, Reference [7] proposes a bi-level planning model for distributed PV-ESS to maximize revenue, while Reference [8] presents a multi-objective planning model to determine the economic capacity of ESS for solar PV support. However, these studies neglect the synergistic potential of coordinated PV and ESS deployment.

Independent planning of PV generation and ESSs fails to adequately

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Table 1
A taxonomy of reviewed papers on distributed PV and ESS planning.

Ref.	Co-planning	PV uncertainty	Accurate probability distribution set	Efficient short-circuit current calculation method	Voltage deviation optimization
[3,4,8,15,19]	×	×	×	×	✓
[5,6,7]	×	×	×	×	×
[10–14]	×	×	×	✓	×
[16–18]	×	✓	×	×	✓
This paper	✓	✓	✓	✓	✓

* √: The item is considered. ×: The item is not considered.

coordinate supply with grid demand. For example, when all PV output is directly fed into the network, the economic motivation for deploying ESSs diminishes significantly. Consequently, ESS installations often remain underutilized, leading to high idle rates and prolonged payback periods. Furthermore, mitigating PV variability requires the grid to hold substantial backup capacity, thereby increasing operational complexity and costs, ultimately compromising system stability.

Techno-economic limitations of ESSs further hinder their effective utilization. During stable grid conditions with low electricity prices, the marginal benefit of operating storage units (especially those with high power ratings and low state-of-charge) is negligible, resulting in suboptimal usage and reduced returns on investment. Therefore, the absence of coordinated PV-ESS planning diminishes the overall PV integration capacity and adversely affects the economic performance of distribution networks.

To overcome these challenges, this paper proposes an integrated PV-ESS configuration that leverages synergies. This coordinated scheme yields several key advantages. First, ESSs can swiftly absorb or inject active and reactive power, enabling dynamic power exchange to improve voltage regulation, mitigate voltage sags, and enhance power quality. Second, due to their "low-SOC, high-C-rate" characteristic, ESSs can buffer PV output fluctuations before grid injection, thereby stabilizing power delivery. Third, co-located ESS units (e.g., battery packs) and PV arrays at a common point of interconnection (PCC) can discharge substantial energy rapidly during faults; this increases the short-circuit current at the PCC and allows circuit breakers to trip promptly, ensuring effective fault isolation. Finally, although integrated PV-ESS planning may increase inverter rating requirements, sharing auxiliary infrastructure (e.g., transformers) can reduce overall costs and improve economic efficiency and enhance load-carrying capability.

In the planning process, two operational constraints—short-circuit current and voltage deviation—are intimately linked to power flow and significantly influenced by PV uncertainty, making them critical determinants of PV hosting capacity. Worldwide, methods for calculating short-circuit currents at the PCC in distribution systems are under continuous development. The IEC 61660 standard [9] offers a method for short-circuit current calculation, but it is ill-suited for modern AC/DC hybrid networks. Reference [10] presents a short-circuit current calculation method based on discrete-time modeling. Reference [11] proposes a Graph Attention Network (GAT)-based model to balance computational precision and speed. Reference [12] performs short-circuit analysis based on a cluster or distributed inverter-based distributed generation (IBDG) to calculate the unbalanced short-circuit current flowing. An affine optimization model is first established to characterize the short-circuit current interval of a transmission line and is constrained in Ref. [13]. Reference [14] proposes a short-circuit current partitioning calculation method considering the degree of voltage drop at the grid-connected point of DG. Nevertheless, PV uncertainty can induce voltage sags at PV-integrated buses, causing dynamic variations in short-circuit currents that existing methods often neglect. This oversight leads to overly conservative planning and diminished economic efficiency in distribution networks.

Furthermore, with large-scale PV integration, traditional "passive" distribution networks are evolving into "active" ones, shifting from unidirectional to bidirectional power flow and increasing voltage deviations. Reference [15] proposes an effective control strategy for voltage regulators to maintain network voltage. Reference [16] presents a hierarchical optimal control framework to support voltage. A bi-level inter-phase coordinated control method for voltage regulation is employed in Ref. [17]. Reference [18] proposes a cost-effectiveness analysis method when combining reactive power compensators and storage batteries. A decentralized control method for coordinating of on-load tap changer (OLTC) transformers and PV inverters is proposed in Ref. [19]. Although the above studies mitigate voltage deviation issues through devices such as converters to enhance network security, the associated construction costs negatively impact economic efficiency.

Table 1 provides a taxonomy of existing planning methods for distributed PVs and ESSs. A comprehensive review of the literature reveals the following research gaps: Current studies fail to harness the synergistic benefits of co-planning for PVs and ESSs, which restricts PV capacity, reduces economic efficiency, and exacerbates voltage fluctuations. Additionally, many previous studies overlook the impact of PV uncertainty on PCC short-circuit current levels, which risks non-compliance with short-circuit constraints and compromising planning schemes under extreme weather conditions. This narrow focus has hindered the coordinated development of PVs and ESSs, highlighting the need for a co-planning approach that integrates PV uncertainty.

To fill this research gap, this paper establishes a co-planning model for distributed PVs and ESSs that incorporates PV uncertainty while embedding short-circuit current constraints. The model aims to optimize planning costs and voltage deviation while enhancing network hosting capacity under security constraints. First, a distributionally robust optimization algorithm is employed to construct a probability distribution set for PV uncertainty, generating both representative and extreme scenarios. Optimal siting and sizing for PVs and ESSs are then determined across various scenarios to minimize economic costs and voltage deviation. Subsequently, the short-circuit current level at PCC is iteratively calculated using the equivalent voltage source method, and an improved C&CG algorithm is proposed to efficiently solve the model.

The innovations of this paper are as follows:

(1) To enhance the hosting capacity of active distribution networks, a SOCP model for co-planning distributed PVs and ESSs is developed. This model integrates the probabilistic distribution of PV uncertainty and incorporates short-circuit current constraints to improve planning rationality and effectiveness.
(2) To address the challenge of solving high-dimensional models, an improved C&CG algorithm that utilizes short-circuit current constraints as cutting planes is proposed. This algorithm iteratively calculates PCC short-circuit current levels by accounting for power flow variations, thereby enhancing computational efficiency without sacrificing accuracy.

This paper is organized as follows. In Section 2, the planning frameworks of PVs and ESSs are analyzed, and a method to address these issues is proposed. Section 3 proposes a method for accurately calculating the level of short-circuit current and voltage deviation. Section 4 proposes a co-planning model considering the probabilistic distribution of PV uncertainty. The verification of the proposed model via case studies is shown in Section 5. Finally, Section 6 summarizes the

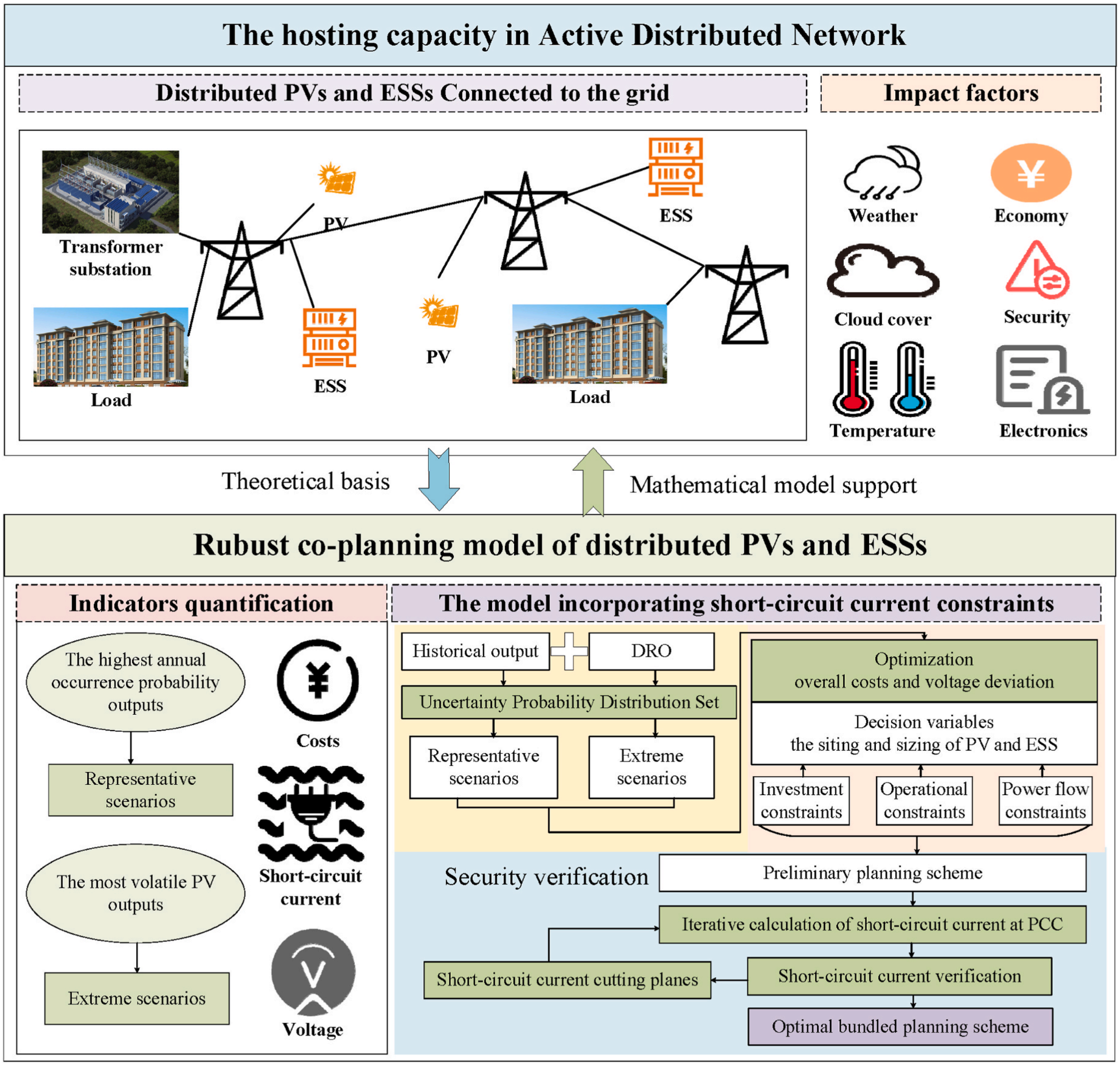


Fig. 1. The co-planning framework of the distributed PVs and ESSs.

conclusions of the study.

2. Co-planning framework of distributed PVs and ESSs

This section proposes the co-planning framework of distributed PVs and ESS at the same bus to achieve stable operation of the distribution network with minimal economic cost.

2.1. Problem description

The factors influencing the hosting capacity of distribution networks can be broadly classified into two main categories:

1) Environmental factors: The output time series of PV generation and load demand are significantly impacted by weather conditions, cloud cover, and seasonal temperature fluctuations. These uncertainties impose substantial constraints on the network's hosting capacity. If planning relies solely on representative scenarios without accounting for extreme cases caused by severe weather, the PV integration results may be overly optimistic, introducing considerable risks to the planning scheme. Conversely, focusing exclusively on extreme scenarios could lead to an underestimation of PV capacity, resulting in overly conservative planning that reduces the economic efficiency of the distribution network.
2) Grid factors: The network's hosting capacity is influenced both by the maximum capacity of power electronics within the network (including transformer types, connection methods, line lengths, and the overall network structure) and by security constraints such as power quality, reliability, and economic efficiency. Power quality is mainly reflected in bus voltage deviations, while safety reliability is linked to short-circuit currents, and economic efficiency is associated with operational costs. Among these factors, short-circuit current and voltage deviations, both closely related to power flow, are crucial elements impacting hosting capacity.

With the large-scale integration of distributed PVs, the short-circuit current characteristics at the PCC in traditional distribution networks have changed significantly [20], and voltage fluctuations have noticeably increased. Due to the limited capacity of power electronics (e.g., PV inverters), the PCC short-circuit current provided by PVs is relatively low, which reduces protection sensitivity and increasing the risk of circuit breaker failure. In the event of a circuit fault, delayed disconnection can jeopardize the protection of other equipment. Additionally, PV generation variability—affected by solar radiation, ambient temperature, and cloud cover—can induce reverse fault currents through the PCC, potentially triggering protection malfunctions and amplifying bus voltage fluctuations, thereby degrading power quality. Furthermore, PV generation instability can lead to inaccuracies in short-circuit current calculations; excessively high short-circuit currents in extreme scenarios may damage electrical equipment. Thus, it is essential to accurately calculate PCC short-circuit current levels, control them within a safe range, and optimize voltage deviations in the distribution network to enhance the network's hosting capacity and reduce economic costs.

2.2. Planning framework

To address these challenges, it is essential to strengthen the distribution network is essential to support large-scale integration of distributed PVs and ESSs. This paper proposes solutions from both planning and modeling perspectives:

1) Planning perspective: A co-planning approach is proposed for collocating PVs and ESSs at the same bus, leveraging their synergy to stabilize PV output fluctuations and enhance network security. The bundled ESSs can quickly absorb or release active and reactive power, thereby improving voltage deviation, reducing voltage sags, and enhancing power quality. Additionally, the ESSs' energy-shifting capabilities help smooth PV output, thereby mitigating the impact of uncertainty. During grid faults, ESSs can rapidly supply energy, boosting the short-circuit current at the PCC to ensure the timely operation of circuit breakers and isolation of faulted circuits. The ESSs can also absorb PV power during voltage sags, balancing power on both sides of the DC bus and suppressing inverter output to maintain stable network operation [21]. Although bundling PVs and ESSs increases inverter costs, the overall economic efficiency and PV capacity of the network improve through shared infrastructure such as transformers.
2) Modeling perspective: Based on a distributionally robust optimization (DRO) algorithm, a probabilistic distribution set for PV uncertainty caused by factors such as solar irradiance is constructed. Scenarios with the most volatile PV outputs are defined as extreme scenarios, while those with the highest annual occurrence probability are defined as representative scenarios. The comprehensive planning for PVs and ESSs across various scenarios enhances the effectiveness of scenario selection. Furthermore, an integrated second-order cone programming (SOCP) model is developed to improve the network's hosting capacity. The model embeds PCC short-circuit current constraints for PVs and ESSs, with the aim of optimize voltage deviation and economic costs under security constraints while enhancing computational efficiency.

The framework for the co-planning of PVs and ESSs, which considers PV uncertainty, is illustrated in Fig. 1. This framework enhances the hosting capacity of the distribution network by generating scenarios based on environmental factors and optimizing indicators that consider grid factors. The co-planning model, which incorporates short-circuit current constraints, is solved iteratively using an enhanced column-and-constraint generation (C&CG) algorithm.

3. Short-circuit current and voltage deviation calculation method

In this section, a method is proposed for accurately calculating two security indicators: short-circuit current and voltage deviation, to ensure the stable operation of the distribution network.

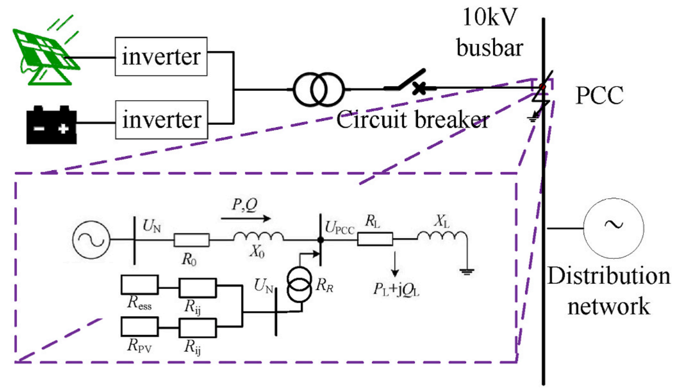


Fig. 2. Equivalent circuit model of distributed PVs and ESSs.

3.1. Short-circuit current calculation considering distribution network topology

When distributed PVs and ESSs connect to the distribution network, they contribute to an increase in the short-circuit current at the PCC. However, traditional PVs and ESSs typically connect to the distribution network through inverters, which are often approximated as current sources [22]. During a short circuit, the current-limiting control of inverters restricts their contribution to the short-circuit current, making it challenging to meet the required short-circuit current constraints at the PCC. To address this, a method is proposed for co-planning distributed PVs and ESS as an equivalent voltage source. By considering the network topology, this approach enables an accurate calculation of short-circuit current levels at the PCC to satisfy security requirements.

Fig. 2 provides a detailed illustration of the equivalent circuit model for distributed PVs and ESSs. As depicted, PVs and ESSs share common infrastructure, such as transformers and circuit breakers, and are connected to the 10 kV bus of the distribution network through a shared AC/DC converter and grid connection point. The power flow at the point of common coupling (PCC) is simplified, with the short-circuit current consisting of contributions from the equivalent voltage source, the controlled current source, and associated equipment, including transformers.

The initial PCC short-circuit current is calculated using the equivalent voltage source method recommended by GB/T 15544 [23]. The initial short-circuit current of PVs is determined based on the PV module conductance and nominal voltage, with the specific formula given in (1). Equation (2) represents the calculation of the equivalent conductance of PVs. The PV access point is denoted as i.

$$I_{i,PV} = \frac{cU_N}{\sqrt{3}} G_{ki,PV} + G_{ki,PV} \sum_{i,j} \frac{\Delta I_{ti,PV}}{G_{ij}} \quad \forall i,j \in B \tag{1}$$

$$G_{ki,PV} = E_{PV} G_{i,PV} \quad \forall i \in B \tag{2}$$

when energy storage is connected to the distribution network, the internal converging lines are equivalent to an equivalent impedance, and each energy storage unit is represented as an equivalent energy storage unit. The equivalent energy storage unit and the equivalent impedance are connected in series, and the short-circuit current output by the equivalent energy storage unit is the sum of the output currents from each energy storage unit, with the initial value calculated as shown in (3). The ESS access point is denoted as j.

$$I_{tj,ess} = \frac{cU_N}{\sqrt{3}} G_{tj,ess} + G_{tj,ess} \sum_{i,j} \frac{\Delta \dot{I}_{tj,ess}}{G_{ij}} \quad \forall i,j \in B \tag{3}$$

$$G_{tj,ess} = E_{ess} G_{j,ess} \quad \forall j \in B \tag{4}$$

The PCC short-circuit current I includes the short-circuit currents

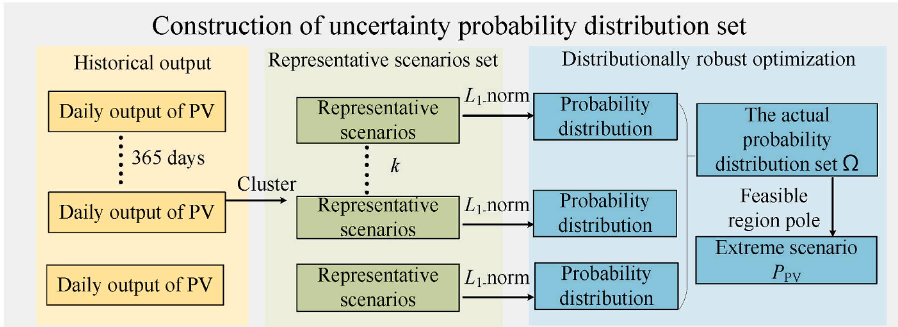


Fig. 3. Implementation procedure of the PV uncertainty set.

injected by the PVs, ESS, and transformers, with the calculation formula shown in (5). The short-circuit current injected by the transformer is calculated based on its short-circuit capacity and impedance percentage, as shown in (6).

$$I_{SCC,l} = I_{tl,PV} + I_{tl,ess} + I_{R,l} \quad \forall l \in B \tag{5}$$

$$I_{R,l} = \frac{S_{SCR}}{U_k\%} \quad \forall l \in B \tag{6}$$

3.2. Calculation of voltage deviation in the distribution network considering power sources

With the large-scale integration of photovoltaic systems, fluctuations in photovoltaic output power caused by factors such as weather and environmental temperature severely impact the power quality of the distribution network. Coordinated planning of energy storage can effectively reduce voltage deviation in the distribution network. When only photovoltaic systems are connected, the normal operating output power of the photovoltaic system will cause voltage fluctuations at the point of common coupling (PCC), with the calculation formula shown in (7).

$$\Delta U = \frac{\Delta P_{PV} R_0 + \Delta Q_{PV} X_0}{U_N^2} \tag{7}$$

The reactive power from photovoltaic systems can be negligible, simplifying (7) as shown in (8).

$$\Delta U = \frac{\Delta P_{PV} R}{U_N^2} \tag{8}$$

when coordinating the configuration of photovoltaic systems and energy storage, the voltage at the point of common coupling (PCC) is calculated as shown in (9):

$$\begin{cases} U_i = U_{g,i} + \dfrac{P_{ess} Z_{ess}}{U_{N,i}^2} \\ Z_{ess} = Z_0 + Z_G \| Z_L \| Z_{PV} \end{cases} \tag{9}$$

4. Distributed PVs and ESS coordinated planning model

This section constructs a coordinated planning model for distributed PVs and ESS using DRO to address the PV uncertainty, aiming to optimize voltage deviation and annual comprehensive costs, while embedding short-circuit current constraints.

4.1. Uncertainty probability distribution set

The output time series of PV generation and load demand are significantly affected by weather conditions, cloud cover, and seasonal temperature variations, which introduce considerable uncertainty. Existing methods for modeling such uncertainty can generally be categorized into two types: scenario-based approaches and probability-driven approaches.

Scenario-based methods generate PV output sequences using a set of representative scenarios. While these methods offer computational efficiency and simplicity, they often fail to accurately capture the stochastic nature of real-world conditions. In contrast, probability-driven methods, such as stochastic optimization and robust optimization, are widely employed for uncertainty modeling. Robust optimization, specifically, aims to determine a solution that remains feasible under worst-case conditions by defining a bounded uncertainty set and solving for the optimal decision at its extreme values. Consequently, the construction of the uncertainty set is critical for balancing the accuracy and computational efficiency of robust optimization models. However, a major limitation of robust optimization is its neglect of the probability distribution of uncertain variables, as it focuses exclusively on worst-case scenarios, which can lead to overly conservative solutions.

To address these limitations, distributionally robust optimization (DRO) has been proposed as a more balanced approach. DRO integrates the probabilistic characteristics of stochastic optimization with the consevativeness of robust optimization by seeking solutions that are optimal under the worst-case probability distribution within a specified ambiguity set. This method mitigates the over-conservatism of traditional robust optimization and reduces dependence on precise probabilistic information, which is often unavailable or difficult to estimate accurately. Therefore, in this study, a probability distribution set representing PV uncertainty is constructed using the DRO framework, to provide a more realistic and flexible modeling approach.

Specifically, based on the historical time series set Z of PV output, k-medoids clustering is used to obtain K scenario sets, with the center of each set defined as a representative scenario $\boldsymbol{S}$, i.e., $\boldsymbol{S} = \{S_1, S_2, \ldots, S_k\}$. PVs and ESS are bundled across different scenarios to improve the universality and rationality of the planning results. As illustrated in Fig. 3, each representative scenario has an annual occurrence probability of $N_s (1 \le s \le k)$. The probability distribution set for these scenarios is $\boldsymbol{P}^0 = [P_1^0, P_2^0 \ldots, P_s^0]$, where the original probability distribution is given by $P_s^0 = N_s / Z$ $(1 \le s \le k)$. Using the L_1-norm, the actual probability distribution set Ω for PV is defined as detailed in equation (10). A distributionally robust optimization algorithm is then applied to establish the uncertain probability distribution set for PV, based on the actual probability. As the parameter γ_1 increases, the fluctuation range of PV characterized by this probability distribution set also expands, enhancing the robustness of the model.

$$\Omega = \left\{ P \middle| \left\| P - P^0 \right\| \le \gamma_1 \right\} = \left\{ p_s \middle| \begin{array}{l} \sum\limits_{s=1}^{k} \left| P_s - {P_s}^0 \right| \le \gamma_1 \\ \sum\limits_{s=1}^{k} p_s = 1 \\ p_s \ge 0, s = 1, 2, \ldots, k \end{array} \right\} \tag{10}$$

Since the scenario with the most significant PV fluctuation occurs at the vertices of the feasible domain, two 0–1 variables, α, and β, are used to describe the upper and lower boundaries of the photovoltaic output, as shown in (11). When $\alpha = 1$, the output reaches the maximum fluctuation value. When $\beta = 1$, the output reaches the minimum fluctuation value.

$$P_{PV} = P_{PVf} + \alpha\gamma_1 - \beta\gamma_1 \tag{11}$$

$$\begin{cases} \alpha + \beta \le 1 \\ \sum(\alpha+\beta) \le \Gamma \end{cases} \tag{12}$$

Equation (12) shows the constraints of the PV fluctuation variables, which are mutually exclusive. Γ limits the number that simultaneously reaches the fluctuation limit. The larger Γ, the more intense the PV fluctuation, and the more extreme scenarios.

The current conventional method achieves planning by determining the variation range of the PV uncertainty parameter, which requires extensive computation and is inefficient. This paper establishes a PV uncertainty probability distribution set based on DRO and generates different scenarios, thus improving the effectiveness of scenarios.

4.2. The co-planning model

A second-order cone programming model with embedded short-circuit current constraints is then established to determine the optimal sizing and siting scheme. This model aims to optimize voltage deviation and overall cost for the distribution network, encompassing both planning and operational expenses. The objective function, defined in equation (13), includes decision variables $\mathbf{x}$, which represent the access buses and capacities of PV and ESS. The occurrence probabilities of different scenarios, all of which are contained within the probability distribution set Ω. The objective function's calculation formula is provided below:

$$f(\mathbf{x}) = \min_{P_s\in\Omega}(C_{inv} + \sigma C_{op} + C_{vol}) \tag{13}$$

Typically, the investment cost includes the cost of the PV system and the ESS. The operation cost is the electricity purchased from the main grid, which is given by:

$$C_{\mathrm{inv}} = \frac{\delta(1+\delta)^y}{(1+\delta)^y - 1}(\rho_{\mathrm{pinv}}P_{ess} + \rho_{\mathrm{einv}}E_{ess}) + C_{PV,inv}E_{PV} + C_{ess,om}E_{ess} \tag{14}$$

$$C_{\mathrm{op}} = \sum_i \rho_{pbuy}P_g \tag{15}$$

$$C_{vol} = \sum_{i=2}^{N} |U_i - U_1| \tag{16}$$

1) Investment constraints of PVs and ESSs

Equations (17) and (18) define the capacity of the PV system and the ESS limits concerning the grid. The capacity constraints of the PV system and the ESS are as follows:

$$E_{PV,\min} \le E_{PV} \le E_{PV,\max} \tag{17}$$

$$E_{ess,\min} \le E_{ess} \le E_{ess,\max} \tag{18}$$

2) Operational constraints of ESS devices

Equations (19) and (20) model the charging and discharging constraints of the ESS and the SOC balance, respectively. Here, the charging and discharging of the ESS are controlled by the variables u_{dch} and u_{ch}. Moreover, (19) indicates that the BESS cannot charge and discharge at the same time. The SOC is calculated from the initial SOC capacity and charge/discharge power per hour. Equation (20) shows that the SOC must be less than or equal to the maximum SOC and greater than or

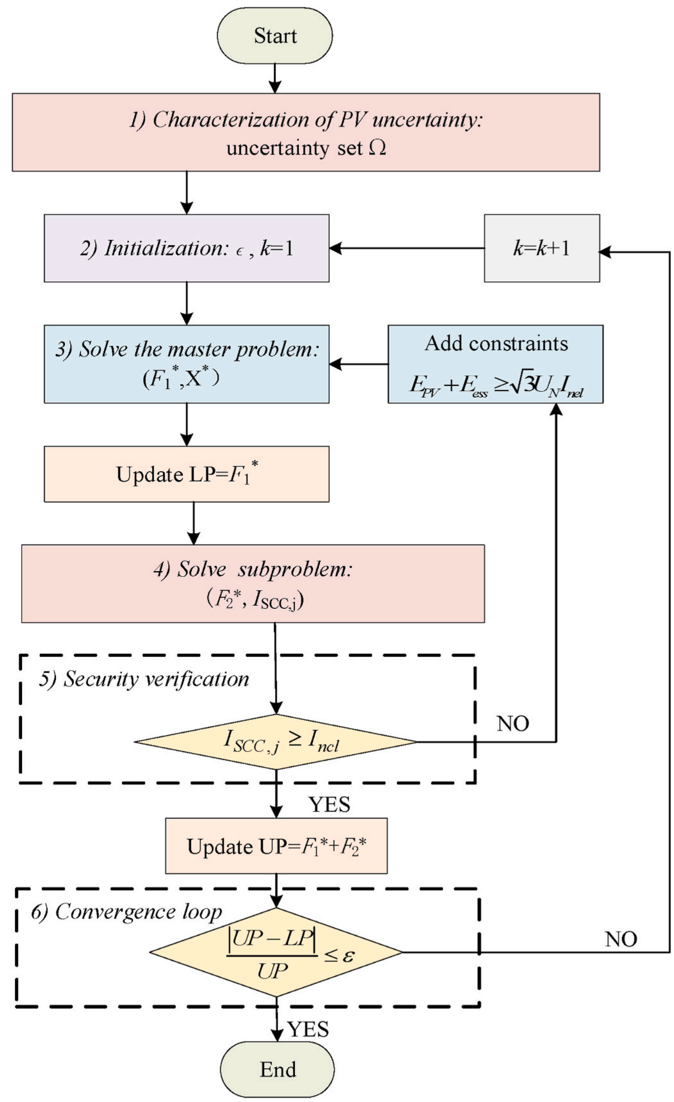


Fig. 4. Workflow of the proposed two-stage coordinated model.

equal to the minimum SOC.

$$\begin{cases} u_{dch}(t) \times P_{dch,\min} \le P_{dch}(t) \le u_{dch}(t) \times P_{dch,\max} \\ u_{ch}(t) \times P_{ch,\min} \le P_{ch}(t) \le u_{ch}(t) \times P_{ch,\max} \quad \forall t \in T \\ u_{dch}(t) + u_{ch}(t) \le 1 \end{cases} \tag{19}$$

$$\begin{cases} SOC_{ess,\min} \le SOC_{ess}(t) \le SOC_{ess,\max} \\ SOC_{ess}(t) = SOC_{ess}(t-1) + \eta_{ch}P_{ch}(t) - \dfrac{P_{dch}(t)}{\eta_{dch}} \quad \forall t \in T \end{cases} \tag{20}$$

3) Power flow constraints

Equation (21) models the constraints of active and reactive power balance. Here, the active power variables include the uncertain output of the PV P_j^{PV}, the active power load P_j^{load}, the battery output power P_j^{ess}. Let $\widetilde{I_{ij}} = I_{ij}^2$ and $\widetilde{U_{ij}} = U_{ij}^2$, the power flow constraints would be as follows:

$$\begin{cases} \sum_{i\in P(j)}(P_{ij} - \widetilde{I}_{ij}r_{ij}) - \sum_{l\in C(j)}P_{jl} = P_j^{load} - P_j^{PV} - P_j^{ess} \\ \sum_{i\in P(j)}(Q_{ij} - \widetilde{I}_{ij}x_{ij}) - \sum_{l\in C(j)}Q_{jl} = Q_j^{load} \end{cases} \quad \forall i,j,l \in B \tag{21}$$

4) Voltage constraint

Equation (22) defines the voltage for bus j, where r_{ij} and x_{ij} represent the resistance and reactance values of branch ij, respectively.

$$\widetilde{U}_j(t) = \widetilde{U}_i(t) - 2\left(P_{ij}(t)r_{(t)ij} + Q_{ij}(t)x_{ij}(t)\right) + \widetilde{I}_{ij}(t)\left(r_{ij}^2(t) + x_{ij}^2(t)\right), \forall i,j \in B \tag{22}$$

$$\underline{U_j^2} \le \widetilde{U}_j(t) \le \overline{U_j^2}, \forall j \in B \tag{23}$$

5) Current constraint

Equation (24) enforces the bound for the current of branch ij.

$$\underline{I_{ij}^2} \le \widetilde{I}_{ij}(t) \le \overline{I_{ij}^2}, \forall i,j \in B \tag{24}$$

$$\dot{I}_{kj,ess} = \begin{cases} \dfrac{-|P_{ref}| - jQ_{ref}}{S_N} I_N & U_{kPCC} \ge U_L \\ \min\left\{\dfrac{-|P_{ref}|}{S_N} I_N, \sqrt{I_{\max}^2 - \min\left\{I_{qref}^2, I_{\min}^2\right\}}\right\} - j\min\left\{I_{qref}, I_{\max}\right\} & U_{kPCC} < U_L \end{cases} \tag{33}$$

6) Second-order cone constraint

Through relaxation deformation, (23) and (24) are transformed into a second-order cone constraint:

$$\left\| \begin{matrix} 2P_{ij} \\ 2Q_{ij} \\ \widetilde{I}_{ij} - \widetilde{U}_j \end{matrix} \right\| \le \widetilde{I}_{ij} + \widetilde{U}_j, \forall i,j \in B \tag{25}$$

4.3. Security verification

The initial planning scheme obtained from the above model is subjected to security verification, primarily considering whether the short-circuit current and the port voltage exceed limits. Using the short-circuit conductance of PVs and ESS and the admittance matrix, the k-th port voltage of bus j is calculated, as shown in (26)-(27).

$$U_{ti,PV}(k) = c_{\min} - \frac{c_{\max} G_k}{G_i} + \sum_{i \in B} \frac{\Delta I_{kti}(k)}{G_i} \quad \forall t \in T \tag{26}$$

$$U_{tj,ess}(k) = c_{\min} - \frac{c_{\max} G_k}{G_j} + + \sum_{j \in B} \frac{\Delta I_{tj,ess}(k)}{G_j} \quad \forall t \in T \tag{27}$$

Based on the obtained port voltage, (28) and (29) respectively calculate the short-circuit current for PVs and ESS.

$$\begin{cases} I_{\mathrm{kq}}(k) = \min\left[K_{\mathrm{Lvrt}}\left(0.9 - U_{ti,PV}(k)\right)I_{\mathrm{N}}, K_{\mathrm{qlim}} I_{\mathrm{N}}\right] & U_k < 0.9 \\ I_{\mathrm{kp}}(k) = \min\left[\sqrt{(K_i I_{\mathrm{N}})^2 - I(k)_{\mathrm{kq}}^2}, \left(P_0 / U_{ti,PV}(k)\right) I_{\mathrm{N}}\right] \end{cases} \tag{28}$$

$$\dot{I}_{kti}(k) = \sqrt{I_{\mathrm{kp}}^2(k) + I_{\mathrm{kq}}^2(k)}\, e^{j\arctan\left(-I_{\mathrm{kq}}(t)/I_{\mathrm{kp}}(t)\right)} \tag{29}$$

After obtaining the k-th short-circuit current component, the fault component of the steady-state short-circuit current at bus j is calculated, as shown in equation (30):

$$\Delta I_{kti}(k) = |I_{kti}(k) - I_{kti}(k-1)| \tag{30}$$

Then, the short-circuit current injected by ESS is calculated after obtaining the steady-state component, considering the impact of its charging and discharging states on the short-circuit current. In the discharging state, the short-circuit current is calculated as shown in (31)-(32).

$$\dot{I}_{kj,ess} = \begin{cases} \dfrac{P_{ref} - jQ_{ref}}{S_N} I_N & U_{kPCC} \ge U_L \\ \min\left\{\dfrac{P_{ref}}{S_N} I_N, \sqrt{I_{\max}^2 - \min\left\{I_{qref}^2, I_{\min}^2\right\}}\right\} - j\min\left\{I_{qref}, I_{\max}\right\} & U_{kPCC} < U_L \end{cases} \tag{31}$$

$$I_{qref} = \frac{K_L(U_L - U_{kPCC})}{U_N} I_N \tag{32}$$

In the charging state, the short-circuit current is calculated according to (33), where Pref is positive when flowing into the system.

Therefore, the fault component of the steady-state short-circuit current at bus j is detailed in (34).

$$\Delta \dot{I}_{tj,ess}(k) = \left|\dot{I}_{tj,ess}(k) - \dot{I}_{tj,ess}(k-1)\right| \quad \forall j \in B \tag{34}$$

The iterative process continues until the difference between $U_{ti,\mathrm{PV}}$ and $U_{tj,\mathrm{ess}}$ in two successive iterations falls below a predefined threshold. Once this convergence criterion is satisfied, the iteration terminates, and both the short-circuit current at the point of common coupling (PCC) and the port voltage are determined.

The initial values of the short-circuit current by PV and ESS are calculated according to (4) and (6).

Based on the principle of short-circuit protection, the short-circuit current of distributed PVs and ESS during a fault should exceed the rated breaking current allowed by the circuit breaker, so that the breaker can quickly disconnect the line to achieve the purpose of short-circuit protection. At the same time, the short-circuit current $I_{\mathrm{SCC},l}$ during a fault must not exceed the breaking capacity of the circuit breaker to ensure that it can function properly and disconnect the circuit during a short circuit. Therefore, based on the requirements for circuit breaker short-circuit protection, a short-circuit current constraint is imposed as (35).

$$I_{ncl} \le I_{SCC,l} \le I_z \quad \forall l \in B \tag{35}$$

4.4. Improved C&CG algorithm

The proposed SOCP model for co-planning of distributed PV and ESS is solved sequentially. To enhance solution efficiency, an improved Column-and-Constraint Generation (C&CG) algorithm utilizing short-circuit current cut sets is introduced. This algorithm operates iteratively: first, the master problem is solved using the SOCP model, followed by the subproblem calculation based on the preliminary planning solution. A cut set for PV and ESS accessible capacity is constructed based on short-circuit current constraints, with the iterative process continuing until model convergence is achieved.

For clarity, the detailed solution procedure of the model is illustrated in Fig. 4. The details are as follows:

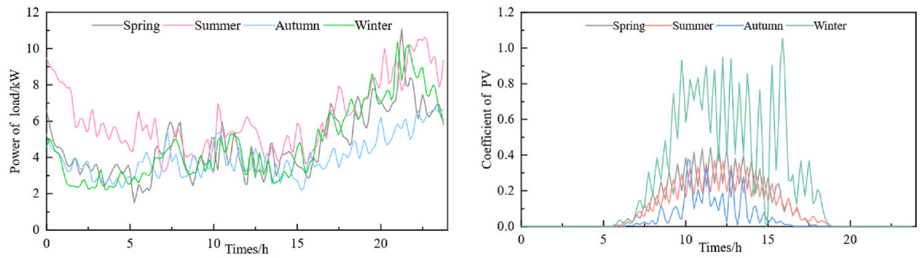


Fig. 5. Time series of output at bus 2.

1) *Characterization of PV uncertainty:* The DRO algorithm is used to construct the PV uncertainty probability distribution set Ω according to equation (10).
2) *Set initial values:* Set the convergence threshold ϵ for the iteration and initialize the number of iterations $k = 1$.
3) *Solving the master problem:* Solve the economic optimization problem F under different scenarios, where P_s represents the extreme scenario in the PV probability distribution set. The linearized objective function of the master problem is detailed in (36). The constraints include equations (17)–(22). The solution of the master problem yields the objective function value F_1* and the preliminary planning scheme X*, as shown in equation (37). The lower bound is updated as LP=F_1*.

$$F = \min_{P_s \in \Omega}\left(C_{inv} + \sigma C_{op} + C_{vol}\right) \tag{36}$$

$$X^{*} = \left\{x_{PV}, x_{ess}, x_{buy}\right\} \tag{37}$$

4) *Solving the subproblem:* Based on scheme X*, the subproblem F_2 is solved using equation (38), yielding the corresponding objective function value F_2*. Simultaneously, the PCC short-circuit current level $I^*_{\mathrm{SCC},l}$ for the distributed PV and ESS is iteratively computed using equations (25)–(33).

$$\Delta E_{ess} = \sqrt{3}U_N\left(I_{nel} - I_{SCC,l}\right) \quad \forall l \in B \tag{38}$$

$$\mathrm{minF}_2 = \frac{\delta(1+\delta)^y(\rho_{\mathrm{einv}}\Delta E_{ess})}{(1+\delta)^y - 1} + C_{ess,om}\Delta E_{ess} \tag{39}$$

5) *Short-circuit current verification:* According to equation (34), verify whether the PCC short-circuit current $I^*_{\mathrm{SCC},l}$ meets the requirements. If the short-circuit current satisfies the circuit breaker protection requirements, update the upper bound as UP= F_1*+F_2*; otherwise, add an ESS capacity constraint using the cutting plane method that considers short-circuit current constraints and add it to the master problem, then return to step 3 to solve again. The cut set formulas are shown in equations (40) and (41).

$$E_{PV} + E_{ess} \geq \sqrt{3}U_N I_{nel} \tag{40}$$

$$P_{PV} = P_{PVf} + \alpha\gamma_1 \tag{41}$$

6) *Convergence loop*: Compare the difference between the upper and lower bounds. If the difference is smaller than the convergence threshold ϵ, the iteration ends, and the planning scheme X* is obtained. Otherwise, set $k = k+1$ and continue iterating until the model converges.

The proposed co-planning model integrates short-circuit current constraint while accounting for PV uncertainties, thereby enhancing the hosting capacity of the distribution network. Practitioners can utilize the proposed method to balance both economy and security, facilitating the optimal siting and sizing of distributed PVs and ESS. This approach improves the accessible capacity of PVs within security constraints and offers a computationally efficient solution. Consequently, this model supports both economic gains and voltage stability, benefiting society with greater reliability and cost-effectiveness.

Table 2
Distributed PVs and ESSs configuration.

Technical Index	Method 1	Method 2	Method 3
Capacity of PV(MW)	3.00	4.80	4.80
Capacity of ESS(MWh)	0	1.94	1.94
Overall costs($\times$ 10^4¥/year)	6141.4	7698	7698
Voltage deviation(p.u./year)	10746	7981.3	6839.7

5. Case study

This section reports the results of the case study. The experiments are based on the distribution network of a 43-bus system in Zhejiang, China, which includes 43 buses, 42 existing lines, 43 loads, and 17 candidate buses for distributed PVs and ESS installations. The set of candidate buses is {2, 4, 7, 13, 15, 18, 19, 20, 21, 23, 26, 28, 29, 31}. We assume that the capacity and location of PVs and ESS are decision variables.

Based on the annual historical output data, the representative scenarios with the highest occurrence probabilities in each of the four seasons are selected, and the extreme scenario with the most significant PV fluctuations in each season is selected. Fig. 5(a) shows the load representative scenarios of bus 2, specifically {S_1, S_2, S_3, S_4}. After constructing the uncertainty probability distribution set for PV using the DRO, the extreme PV scenarios were obtained in Fig. 5(b).

All parameters of the planning model are listed in Tab. A 2. All tests are run using Gurobi 10.0.3 on an Intel(R) Core (TM) i5-8250U CPU at 1.60 GHz.

5.1. Optimal planning scheme of distributed PVs and ESSs

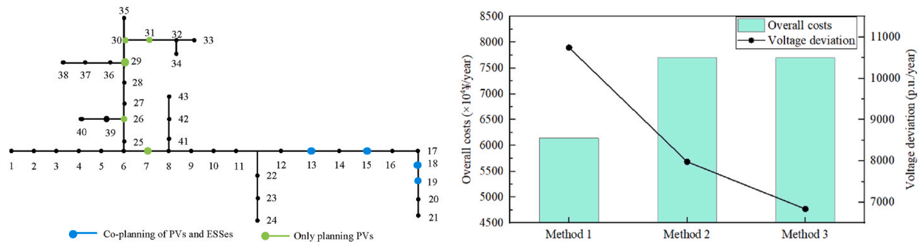
The proposed co-planning method for distributed PVs and ESSs is compared with two alternative planning methods to better demonstrate the superiority of this coordinated approach. The daily PV outputs are recorded at 15-min intervals, resulting in a total of 96 sample points.

Method 1: Only distributed PVs are considered without ESS [24].

Method 2: Distributed PVs and ESSs are planned separately [25].

Method 3: The proposed co-planning model is applied considering PV uncertainty.

By solving the models, the economically optimal planning solutions for the representative scenarios are obtained. The economic performance of the different planning methods is compared, as shown in Table 2. Lines 2–3 in Table 2 display the access capacities of PVs and ESSs, while lines 4–5 present the annual comprehensive costs and the total voltage deviation. As illustrated in Table 2, Method 3 results in a



(a) Co-planning scheme (b) Planning results

Fig. 6. Distributed PVs and ESSs planning scheme in three methods.

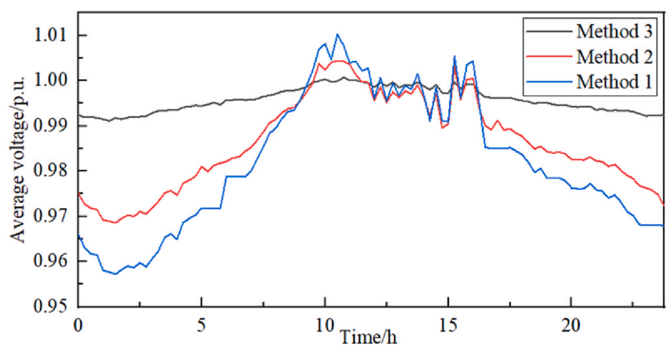


Fig. 7. Comparison of average voltage at distribution network buses.

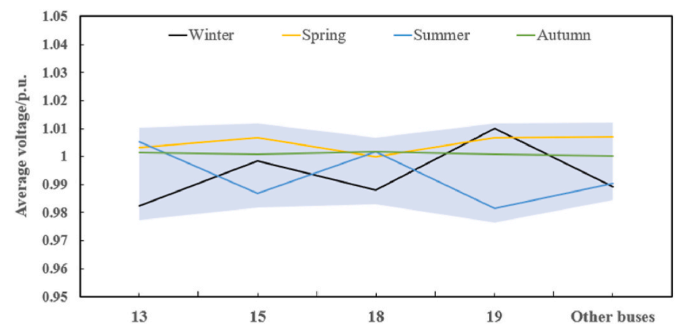


Fig. 8. Buses average voltage in extreme scenarios.

Table 3
Comparison of average short-circuit current.

Buses	13	14	15	16	18	19
Method 2	4.82	3.535	4.82	4.124	4.82	3.623
Method 3	8.13	/	8.43	/	8.19	/
I_{nel}	6	6	6	6	6	6

25.9 % increase in the total capacity of PVs and ESSs compared to Method 1, along with an 11 % reduction in economic costs and a 20 % decrease in voltage deviation. In comparison to Method 2, although Method 3 maintains the same PV and ESS access capacity, it reduces voltage deviation by 16.7 %, highlighting its effectiveness in improving voltage stability.

Fig. 6(a) displays the coordinated configuration of distributed PVs and ESSs obtained using Method 3. The x-axis represents the candidate buses, while the y-axis represents the accessible capacity. Buses 13, 15, and 18 are selected for PV and ESS placement. To further emphasize the superiority of Method 3, Fig. 6(b) compares the economic cost and voltage deviation of the planning solutions. The x-axis represents different planning methods, the left y-axis represents the annual comprehensive economic cost, and the right y-axis represents the voltage deviation. As shown in Fig. 6(b), the planning scheme obtained by Method 3 achieves the lowest economic cost and the smallest voltage deviation in the distribution network.

Fig. 7 illustrates the average voltage changes at the buses. The results demonstrate that the coordinated configuration of PVs and ESSs in Method 3 significantly reduces average voltage fluctuations. This improvement is attributed to the ESS effectively smoothing out PV output fluctuations before being connected to the grid, thereby reducing voltage instability.

The results further demonstrate that the proposed method exhibits superior performance in improving economic efficiency and maintaining voltage stability. When only PVs are planned, the distribution network must rely heavily on electricity purchases from the main grid to meet load demands and mitigate PV fluctuations, resulting in high procurement costs. By co-planning ESSs, PV fluctuations are smoothed, and the dependence on costly grid electricity is substantially reduced, greatly boosting both economic efficiency and voltage stability. Moreover, compared to planning PVs and ESSs individually, co-planning enables ESSs to smooth out fluctuations before grid connection, thus maintaining voltage stability. It also promotes the sharing of infrastructure (such as substations), further reducing investment costs. Consequently, the co-planning of distributed PVs and ESSs not only optimizes economic efficiency and voltage stability but also maximizes the accessible PV capacity.

5.2. Short-circuit current calculation

After obtaining the optimal planning scheme of distributed PVs and ESSs, the short-circuit current at the PCC is further verified to ensure security. To streamline the calculations, this section computes the PCC short-circuit current using the proposed method. The circuit breaker model used for PVs and ESS is the TGB2D-80RW, while the circuit breaker for the transformer connection is the ZN12-10/1250. The transformer has a rated capacity of 100 kVA and a short-circuit impedance of 4 %. The specific parameters required for short-circuit current calculations are provided in Tab. A 3.

Table 3 compares the impact of co-planning versus separate planning (with the same capacity) on the short-circuit current at the PCC. In Method 2, ESSs are independently planned at buses 14, 16, and 19, while PVs are independently planned at buses 13, 15, and 18. In Method 3, PVs and ESS are co-planned together at buses 13, 15, and 18. As shown in Tables 3 , when PVs and ESSs are independently planned, the average PV short-circuit current at the PCC fails to reach the rated breaking current of the circuit breaker, leading to a delayed response during a short-circuit fault and hindering rapid fault clearance. Furthermore, the average short-circuit current of ESSs at the PCC is too low to provide effective short-circuit protection. In contrast, the co-planning of PVs and ESSs appropriately increases the average short-circuit current to levels above the rated breaking current yet within the interrupting current limit. During a short-circuit fault, the proposed co-planning method ensures that the current rapidly reaches the breaker's set threshold,

Table 4
Comparison of calculation efficiency.

Solution	Required time(s)	Improved efficiency (%)
Improved C&CG	107	/
Direct solution	1348	92.06
C&CG	348	69.25

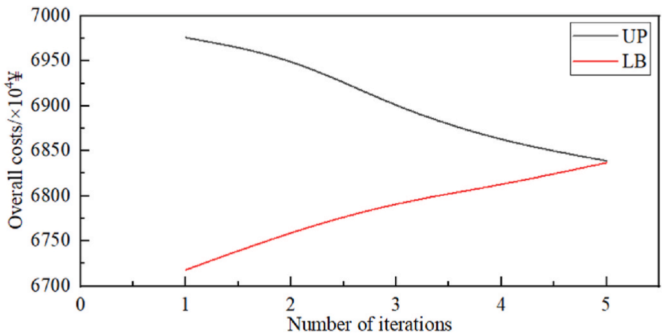


Fig. 9. Iteration number of C&CG.

enabling prompt fault clearance. This approach satisfies short-circuit current constraints and ensures the stable operation of power electronics equipment. Additionally, the co-planning scheme effectively reduces the average short-circuit current at non-connection buses, thus preventing unnecessary losses caused by equipment inadvertent operation.

To further illustrate the impact of voltage deviations across different buses under extreme scenarios, Fig. 8 shows voltage variations at each bus during a fault. The blue-shaded area represents voltage fluctuations due to PV uncertainties. The results indicate that during faults, voltage fluctuations are relatively moderate in the spring and autumn, but more pronounced in the summer and winter. However, all bus voltages remain within the permissible range of 0.95–1.05 p.u.

In summary, the co-planning method proposed in this paper significantly enhances economic efficiency while optimizing voltage deviation and adhering to short-circuit current constraints. Even at the most vulnerable buses during a fault, voltage fluctuations remain within permissible limits, ensuring stable operation. This planning model offers an optimal scheme that effectively balances economic efficiency and security.

5.3. Efficiency and convergence analysis

To clearly demonstrate the improvement in computational efficiency achieved by the enhanced column-and-constraint generation (C&CG) algorithm, a comparative analysis was conducted against the scenario-based direct solution model [26] and the traditional C&CG algorithm [27].

As shown in Table 4, the solution times for the different methods indicate that the improved C&CG algorithm—by incorporating short-circuit current constraints as cutting planes—significantly improves computational efficiency while effectively addressing the uncertainty associated with PV generation. In contrast, the direct solution method relies on a large number of representative scenarios, which substantially increases the computational burden and reduces the overall model efficiency. Although the traditional C&CG algorithm is more efficient than the direct approach, it requires multiple iterations of repeated cuts to converge to the optimal solution. Furthermore, it does not guarantee that short-circuit current constraints at the point of common coupling (PCC) will be satisfied, which compromises the secure operational capacity of the distribution system.

Fig. 9 illustrates the convergence behavior of the proposed algorithm. Both the master problem and the subproblem converge within five iterations, demonstrating strong convergence characteristics and high computational efficiency.

5.4. Sensitivity analysis

To study the impact of different investment parameters on the planning of distributed PVs and ESSs, we analyze the planning results

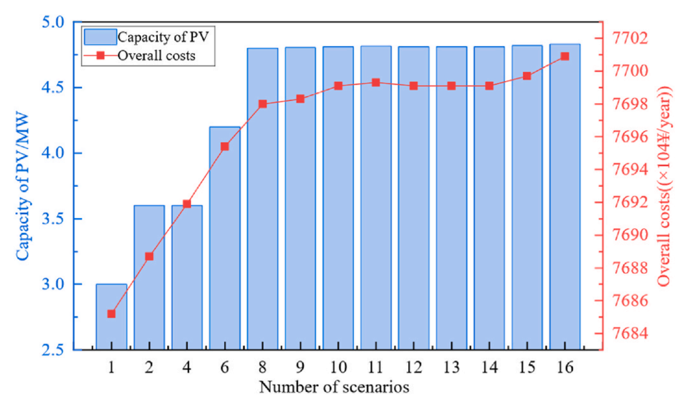
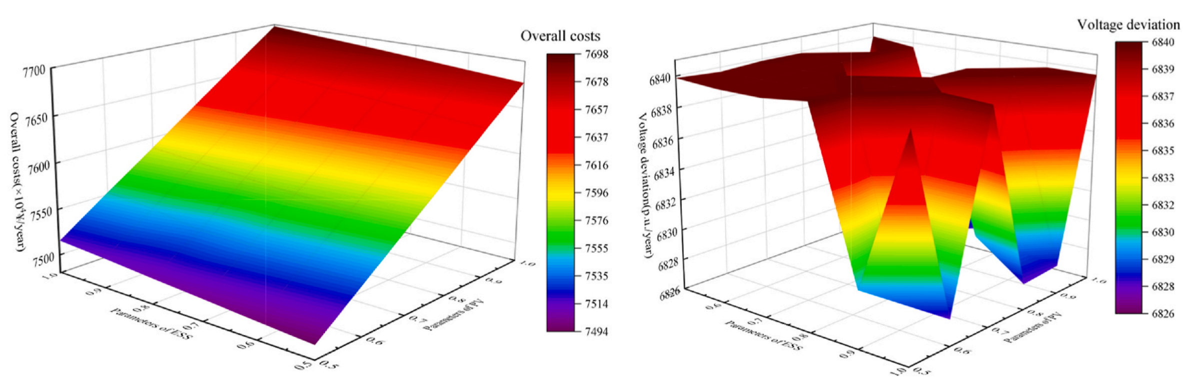


Fig. 11. Sensitivity analysis of number of scenarios.



(a) Overall costs (b) Voltage deviation

Fig. 10. The costs/voltage deviation with different parameters of PV and ESS.

under varying unit investment costs for both PV and ESS. Specifically, given that unit construction costs for PV and ESS are expected to decrease with technological advancements, we use the parameters shown in Tab. A 2 as the baseline and vary the relative costs of PV and ESS from 0.5 to 1 times the baseline value. The results are shown in Fig. 10. As shown, a reduction in the unit investment costs for both PV and ESS significantly reduces the overall economic cost of the planning scheme. In particular, for distributed PV, a 50 % reduction in unit investment costs leads to a 2.64 % reduction in overall costs, which underscores the important role of technological advancements in improving the system's economic efficiency. The voltages in the distribution system remains stable, with no noticeable deviations.

In addition, this study uses a PV uncertainty modeling method based on generating an uncertainty set through representative scenarios. For the k-medoids clustering technique, the number of clusters is a crucial capacity planning parameter that must be carefully selected for capacity planning. Therefore, a sensitivity analysis of the number of scenarios for PV and ESS planning was conducted, with the results presented in Fig. 11. It can be observed that as the number of scenarios approaches 8, both the PV planning capacity and the economic cost of the plan converge. Specifically, when the number of representative scenarios exceeds 8, the PV planning capacity and overall costs stabilize even with further increases in the number of representative scenarios. However, increasing the number of representative scenarios also increases computational time. Consequently, in this study, the number of representative scenarios is determined by balancing overall costs and computational time.

6. Conclusion

In this paper, a two-stage optimization model is proposed to address the co-planning of distributed PVs and ESSs under uncertainty. The first stage determines the optimal investment decisions, including the sizing and siting of PVs and ESSs, within a distributionally robust optimization (DRO) framework that explicitly accounts for PV generation uncertainty. Building on these investment outcomes, the second stage develops operational strategies across a set of generated scenarios, each incorporating short-circuit current constraints to ensure network security under varying operating conditions. The model is iteratively refined by updating the uncertainty set and the associated operational responses until convergence criteria are met, thereby ensuring both robustness and feasibility. By effectively integrating planning and operational decision-making and capturing the nonlinear characteristics of fault currents through a second-order cone programming (SOCP) formulation, the proposed approach significantly enhances system performance. The solution is efficiently obtained using an enhanced column-and-constraint generation (C&CG) algorithm, which utilizes short-circuit current constraints as cutting planes. Nevertheless, the scope of this study does not exhaustively cover all aspects of the proposed methodology, as the SOCP model relies on scenario-based analysis. Consequently, future research should focus on the development of advanced techniques for identifying optimal representative scenarios, thereby enhancing both the accuracy and efficacy of scenario selection.

CRediT authorship contribution statement

Yingzi Wu: Writing – original draft, Visualization, Software, Methodology, Investigation, Conceptualization. **Yuwei Chen:** Writing – original draft, Software, Methodology, Investigation. **Zhiyi Li:** Writing – review & editing, Validation, Supervision, Resources, Methodology. **Sajjad Golshannavaz:** Writing – review & editing, Validation, Supervision, Methodology.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A

Tab. A 1
Variables and implications in Appendix A

Varibles	Implications	Varibles	Implication
Ω	Set of PV uncertainty probability distribution.	U_1	Voltage of root bus.
P	Set of buses connected with PV.	$E_{\mathrm{PV,min/max}}$	Limits of photovoltaic capacity.
E	Set of buses connected with ESS.	$E_{\mathrm{ess,min/max}}$	Limits of ESS capacity.
B	Set of distribution network buses.	$P_{\mathrm{dch,min/max}}$	Power limits of ESS discharge.
$\boldsymbol{\Gamma}$	Set of PV uncertainty.	$P_{\mathrm{ch,min/max}}$	Power limits of ESS charge.
s	The s-th scenario.	$SOC_{\mathrm{ess,min/max}}$	SOC limits of ESS discharge.
t	The t-th period.	$\eta_{\mathrm{dch/ch}}$	Charge and discharge efficiency.
i	The i-th bus.	$u_{\mathrm{dch/ch}}$	Charge and discharge ratios.
j	The j-th bus.	P_{ij}	Active power of branch ij.
l	The l-th bus.	Q_{ij}	Reactive power of branch ij.
k	The k-th iteration.	r_{ij}	Resistance of branch ij.
$f(\mathbf{x})$	Objection function for investment $\mathbf{x}$.	x_{ij}	Inductance of branch ij.
$R_{0/\mathrm{L}}$	The resistance of the transmission line and the connected load.	$\underline{I_{ij}^2}, \overline{I_{ij}^2}$	Limits of the square of the current.
$X_{0/\mathrm{L}}$	The reactance of the transmission line and the connected load.	$\underline{U_j^2}, \overline{U_j^2}$	Limits of the square of the voltage.
R_{PV}	The equivalent resistance of PV.	U_j	The voltage of root bus j.
R_{ess}	The equivalent resistance of ESS.	P_j^{load}	Load active power of bus j.
R_{R}	The equivalent resistance of the transformer.	Q_j^{load}	Load reactive power of bus j.
c	The voltage coefficient.	P_j^{PV}	Active power of the PV system of bus j.
U_n	The nominal voltage.	P_j^{ess}	Active power of ESS of bus j.

(continued on next page)

Tab. A 1 (*continued*)

Varibles	Implications	Varibles	Implication
$G_{\mathrm{kPV/kBESS},j}$	The equivalent short-circuit conductance at bus *j* of PV or ESS.	$P_{\mathrm{PV,f}}$	Predicted value of the PV outputs.
G_{ij}	The magnitude of the admittance matrix of branch *ij*.	$P_{\mathrm{load,f}}$	Predicted value of the load outputs.
$\Delta I_{ti,PV/tj,ess}$	The current increment before and after the fault at bus *j* of PV or ESS.	$c_{\mathrm{max/min}}$	Limits of voltage coefficients.
$G_{i,PV/j,ess}$	The per unit value unit conductance of PV or ESS.	ΔI_{kti}	The *k*-th fault component of steady-state short-circuit current of bus *i* for PV.
$I_{R,l}$	The short-circuit current by the transformer of bus *l*.	$\Delta I_{ktj,ess}$	The *k*-th fault component of steady-state short-circuit current of bus *j* for ESS.
S_{SCR}	The short-circuit capacity of the transformer.	$I_{\mathrm{kq/kp}}$	The reactive/active component of PV short-circuit current.
U_k%	The impedance percentage of the transformer.	K_{Lvrt}	The reactive current coefficient of PV for low voltage ride-through.
$\Delta P_{\mathrm{PV/esss}}$	The increments of active power by PV or ESS.	K_{qlim}	The maximum value of the reactive component of PV.
ΔQ_{PV}	The increments of PV reactive power.	K_{i}	The maximum overload coefficient of PV.
$U_{g,i}$	The voltage of bus *i*.	S_N	Rated capacity of converter.
$Z_{\mathrm{ess/PV}}$	The equivalent impedance of ESS and PV.	I_N	Rated current of converter.
Z_0	The unit impedance of ESS.	I_{max}	Maximum output current of ESS.
$Z_{\mathrm{G//L}}$	The impedance of grid and load.	P_{ref}/Q_{ref}	Reference active/reactive power before the fault.
P_s	The actual probability distribution of *s*-th scenario.	K_{L}	The reactive current coefficient of ESS for low voltage ride-through.
γ_1	The uncertainty range of PV.	U_k	The fault voltage of ESS.
P_{PV}	The actual output curve of PV.	U_{L}	The voltage threshold for entering the low voltage ride-through state.
$C_{inv/op}$	Annual investment cost and operation cost.	I_{ncl}	Rated breaking current of the circuit breaker.
C_{vol}	Voltage deviation.	I_{z}	The interrupting current of the circuit breaker.
P_{g}	Electricity purchased from the main grid.	σ	Number of operating days.
$\rho_{\mathrm{pinv/einv}}$	Investment cost per unit of ESS power capacity and energy capacity.	δ	Discount rate.
ρ_{pbuy}	Unit price of electricity purchased from the grid.	y	Service life of storage.
$C_{\mathrm{ess,om}}$	Operation and maintenance costs of ESS.	$E_{\mathrm{ess/PV}}$	Capacity of the ESS or PV system.
$C_{\mathrm{PV,inv}}$	Construction cost of PVs.	$P_{\mathrm{ess/PV}}$	Power of the ESS or PV system.

All parameters of the co-planning model are listed in Tab. A 2:

Tab. A 2
Basic parameters of the model

Parameters	Values
δ	0.95
ρ_{pinv} (¥/kW)	3600
ρ_{einv} (¥/kWh)	1800
$C_{\mathrm{PV,inv}}$ (¥/kW)	2940
$C_{\mathrm{ess,om}}$ (¥/kW)	600
ρ_{g} (¥/kW)	250 (0:00–6:00)
	750 (6:00–12:00)
	500 (12:00–17:00)
	750 (17:00–19:00)
	250 (19:00–24:00)
e (¥/t)	377
θ	0.97

The specific parameters required for short-circuit current calculations are provided in Tab. A 3.

Tab. A 3
Parameters of short-circuit current

Indicators	Values
c_{min}, c_{max}	0.9,1.05
K_{Lvrt}	1.5
K_{qlim}	1.05
K_{i}	1.1
I_{nel}(kA)	6
I_{z}(kA)	10.4
G_{Bess}(S•MW^{-1})	27.6
G_{PV}(S•MW^{-1})	2.54
Z_{G}(Ω•MW^{-1})	0.5079
Z_{L}(Ω•MW^{-1})	0.1302
Z_0(Ω•MW^{-1})	0.036
Z_{PV}(Ω•MW^{-1})	0.39

References

[1] H. Liu, R. Cao, J. Han, et al., Research status and future prospect of distributed generation hosting capacity assessment of distribution network, Autom. Electr. Power Syst. 1–13 (2024-08-07).

[2] P. Siratarnsophon, K.W. Lao, D. Rosewater, et al., A voltage smoothing algorithm using energy storage PQ control in PV-Integrated power grid, IEEE Trans. Power Deliv. 34 (6) (2019) 2248–2250.

[3] K. Prakash, M. Ali, M.A. Hossain, et al., Planning battery energy storage system in line with grid support parameters enables circular economy aligned ancillary services in low voltage networks, Renew. Energy 201 (Part 1) (2022) (ISSN 0960-1481) 802-820.

[4] S.K. Wankhede, P. Paliwal, M.K. Kirar, Bi-Level multi-objective planning model of solar PV-Battery storage-based DERs in smart grid distribution system, IEEE Access 10 (2022) 14897–14913.
[5] T. Liu, J. Chen, W. Zhang, et al., Joint planning for PV-SESS-MESS in distribution network towards 100% self-consumption of PV via consensus-based ADMM, J. Energy Storage 90 (Part A) (2024) 111747. ISSN 2352-152X.
[6] H.M.A. Ahmed, H.F. Sindi, M.A. Azzouz, et al., Optimal sizing and scheduling of Mobile energy storage toward high penetration levels of renewable energy and fast charging stations, IEEE Trans. Energy Convers. 37 (2) (2022) 1075–1086.
[7] S. Yang, X. Wang, Y. Yang, et al., Bi-level planning model of distributed PV-energy storage system connected to distribution network under the coordinated operation of electricity-carbon market, Sustain. Cities Soc. 89 (2023) (ISSN 2210-6707) 104347.
[8] G. Memarzadeh, F. Keynia, A new hybrid CBSA-GA optimization method and MRMI-LSTM forecasting algorithm for PV-ESS planning in distribution networks, J. Energy Storage 72 (Part D) (2023) 108582. ISSN 2352-152X.
[9] S. Skok, A. Marusic, S. Tesnjak, Transient short-circuit currents in auxiliary DC installations in power plants and substations. 2003 IEEE Bologna Power Tech Conference Proceedings, 2003. Bologna, Italy.
[10] Y. Liu, M. Huang, X. Zha, et al., Short-circuit current estimation of modular multilevel converter using discrete-time modeling, IEEE Trans. Power Electron. 34 (1) (2019) 40–45.
[11] R. Ye, H. Wang, Y. Ma, et al., Enhancing short-circuit current calculation in active distribution networks through fusing superposition theorem and data-driven approach, Int. J. Electr. Power Energy Syst. 161 (ISSN 0142–0615) (2024) 110196.
[12] M.A. Khan, A. Haque, V.S.B. Kurukuru, Dynamic voltage support for low-voltage ride-through operation in single-phase grid-connected photovoltaic systems, IEEE Trans. Power Electron. 36 (10) (2021) 12102–12111.
[13] Y. Zhang, B. Du, B. Pan, et al., An algorithm for short-circuit current interval in distribution networks with inverter type distributed generation based on affine arithmetic, Energy Eng. J. Assoc. Energy Eng. 121 (7) (2024) (ISSN 0199-8595) 1903-1920.
[14] W. Wang, Q. Shao, S. Wang, et al., Partitioning calculation method of short-circuit current for high proportion DG access to distribution network, Energy Eng. J. Assoc. Energy Eng. 121 (9) (2024) 2569–2584 (ISSN 0199-8595.
[15] A. Bedawy, N. Yorino, K. Mahmoud, et al., Optimal voltage control strategy for voltage regulators in active unbalanced distribution systems using multi-agents, IEEE Trans. Power Syst. 35 (2) (2020) 1023–1035.
[16] G.E. Mejia-Ruiz, M. R, et al., Fast hierarchical coordinated controller for distributed battery energy storage systems to mitigate voltage and frequency deviations, Appl. Energy 323 (ISSN 0306–2619) (2022) 119622.
[17] F. Rezaei, S. Esmaeili, A bi-level inter-phase coordinated control method for voltage regulation in unbalanced LV distribution networks using PV-battery energy storage systems, J. Energy Storage 101 (Part B) (2024) (ISSN 2352-152X) 113943.
[18] M. Takagi, K. Fukushima, N. Tagashira, et al., Cost-effectiveness analysis method for voltage stabilization in case of combining storage battery and reactive power compensator, J. Energy Storage 93 (ISSN 2352–152X) (2024) 112065.
[19] M.R. Jafari, M. Parniani, M.H. Ravanji, Decentralized control of OLTC and PV inverters for voltage regulation in radial distribution networks with high PV penetration, IEEE Trans. Power Deliv. 37 (6) (2022) 4827–4837.
[20] K. Li, Q. Huang, R. Fan, et al., Experimental verification of the impact of high proportion distributed photovoltaic access on distribution network protection, Electrotech. Appl. 42 (12) (2023) 16–24.
[21] B. Deng, Y. Wen, L. Yu, et al., Sort-circuit current suppression strategy for receiving-end power grid based on coordination of current limiter configuration and network structure optimization, Autom. Electr. Power Syst. 48 (9) (2024) 151–161.
[22] S. Favuzza, M. Mitolo, S. Moradi, et al., A general methodology for short-circuit calculations in hybrid AC/DC microgrids, IEEE Trans. Ind. Appl. 59 (3) (2023) 2742–2749.
[23] Q/GDW 10738—2020[S]. Beijing, The Guide of Planning and Design of Distribution Network, 2020-12-31.
[24] J. Liu, K. Sun, Z. Ding, et al., Multi-stage planning of distribution network with high penetration renewable energy considering reliability index, IEEE Trans. Ind. Appl. 60 (2) (2024) 2344–2356.
[25] S. Wang, G. Geng, Q. Jiang, et al., Generation expansion planning considering discrete storage model and renewable energy uncertainty: a bi-interval optimization approach, IEEE Trans. Ind. Inf. 19 (3) (2023) 2973–2983.
[26] X. Cao, T. Cao, F. Gao, et al., Risk-averse storage planning for improving RES hosting capacity under uncertain siting choices, IEEE Trans. Sustain. Energy 12 (4) (2021) 1984–1995.
[27] H. Zeng, et al., Data-driven robust planning for distribution network based on extreme scenarios. 2022 Power System and Green Energy Conference (PSGEC), 2022, pp. 739–743. Shanghai, China.